

DSA1101

Introduction to Data Science

Week 7 Tutorial 5

TODAY'S AGENDA

1. Quick Revision
2. Off-site Questions Discussion
3. (If time permits) Discussion of
AY25/25 Sem 1 Midterms Paper

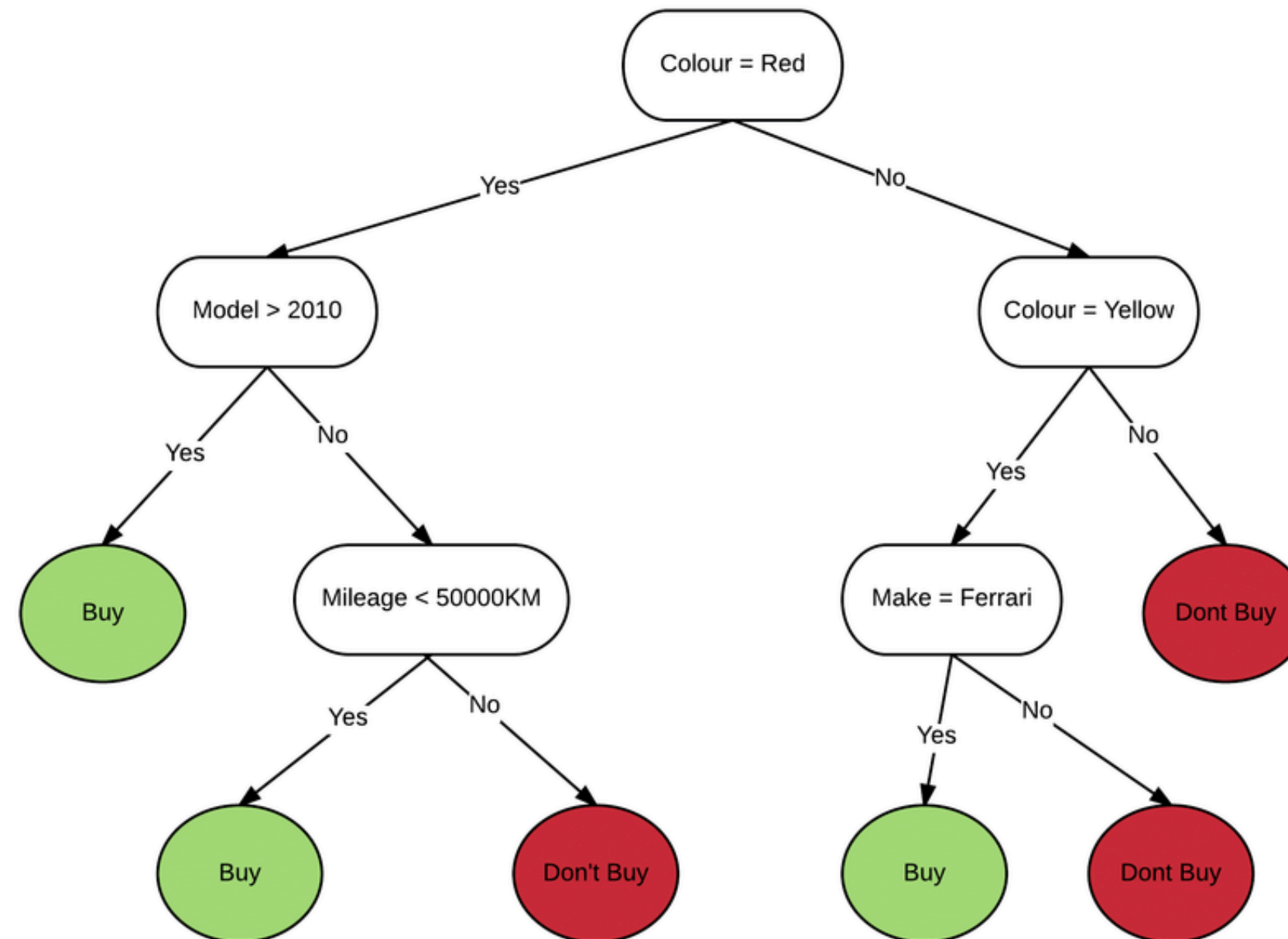
**Going back to school
after break**



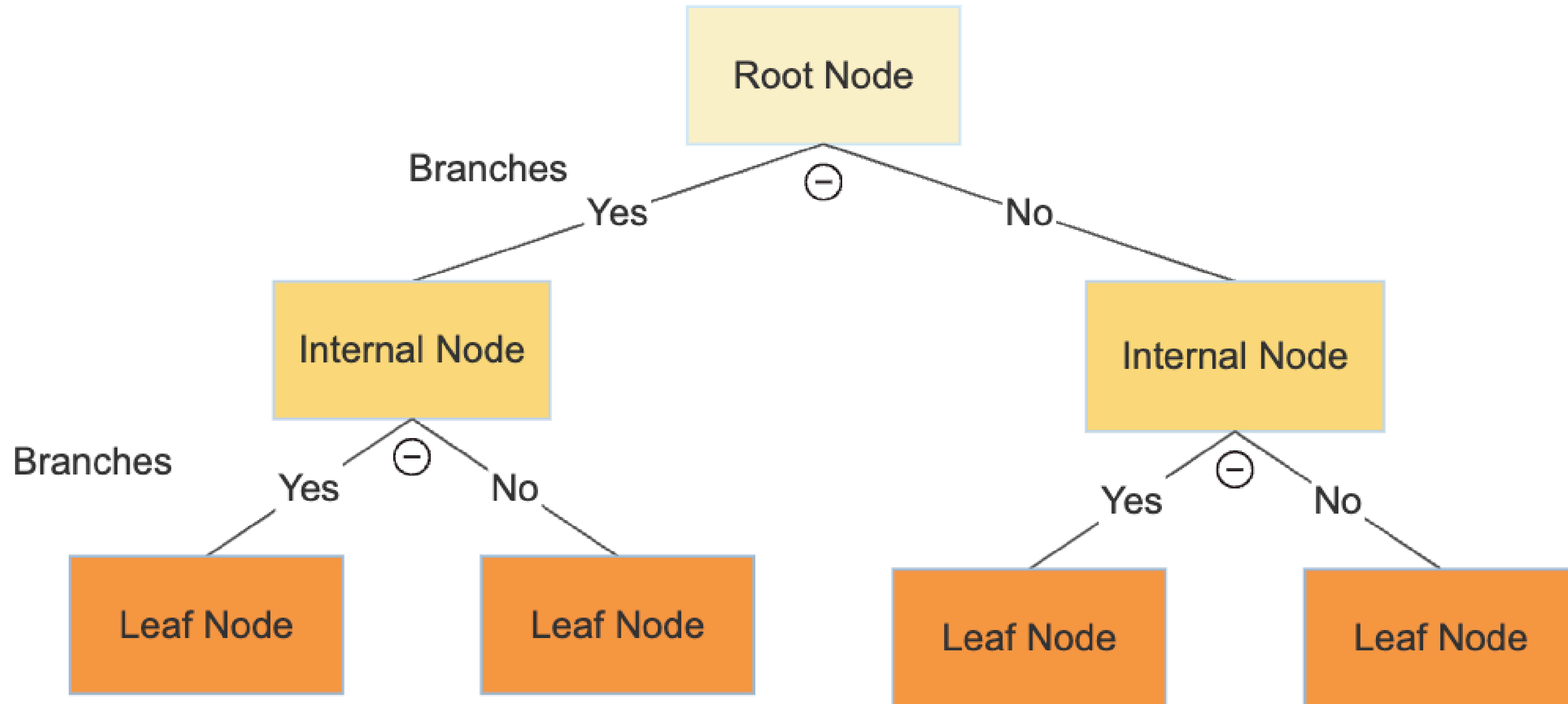
QUICK REVISION

Decision Tree

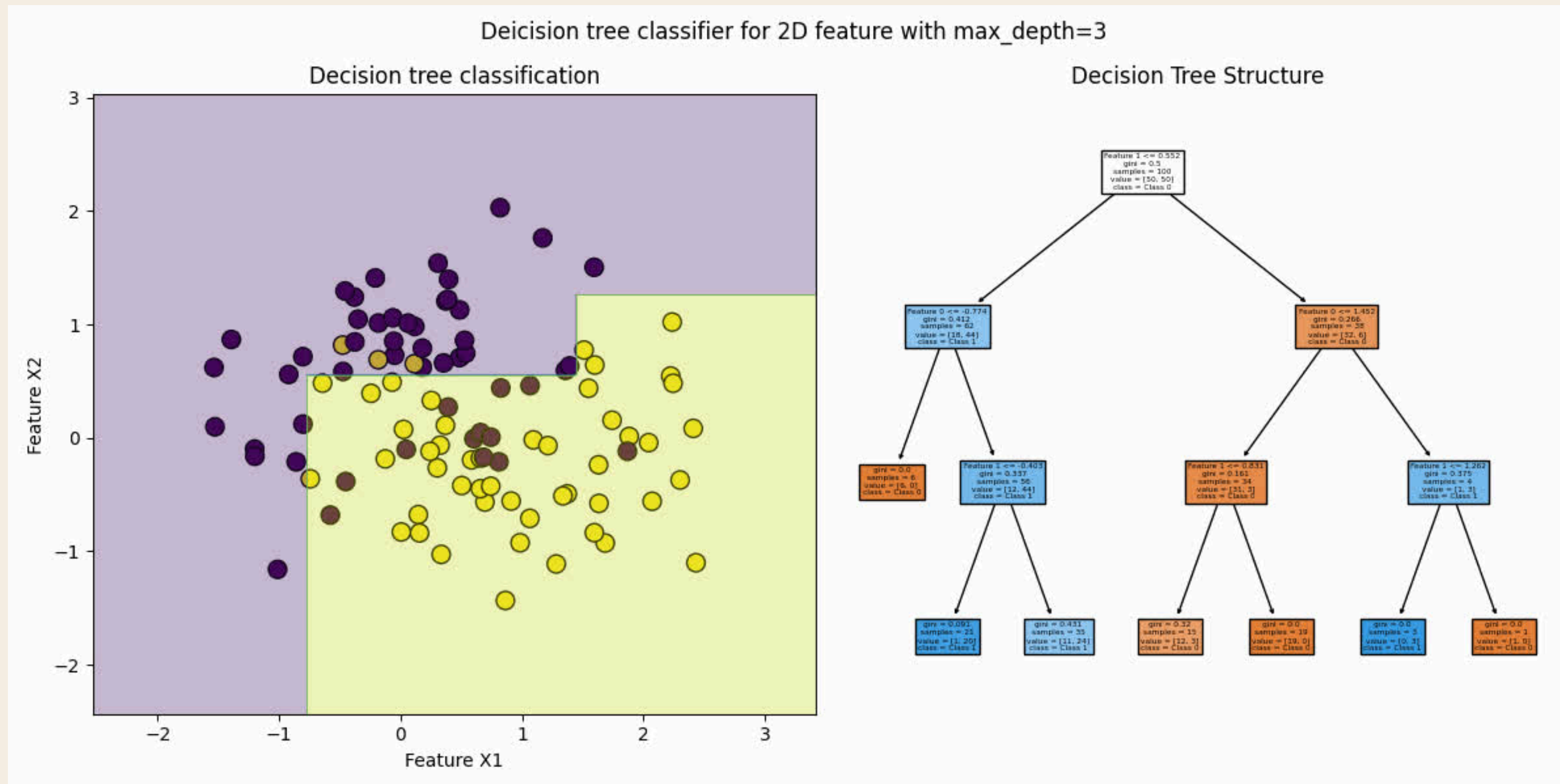
a flowchart-like diagram used to map out decisions and their potential outcomes,
acting as a visual "if-then-else" model



Decision Tree Elements



Decision Tree



(EXTRA) Q: Based on the boundaries, do you think this is overfitting or underfitting?

How does Decision Tree work

High-level idea

1. Start with all data at the root node.
2. Try splitting on every feature.
3. Choose the split that improves purity the most.
4. Repeat recursively until stopping criteria are met.

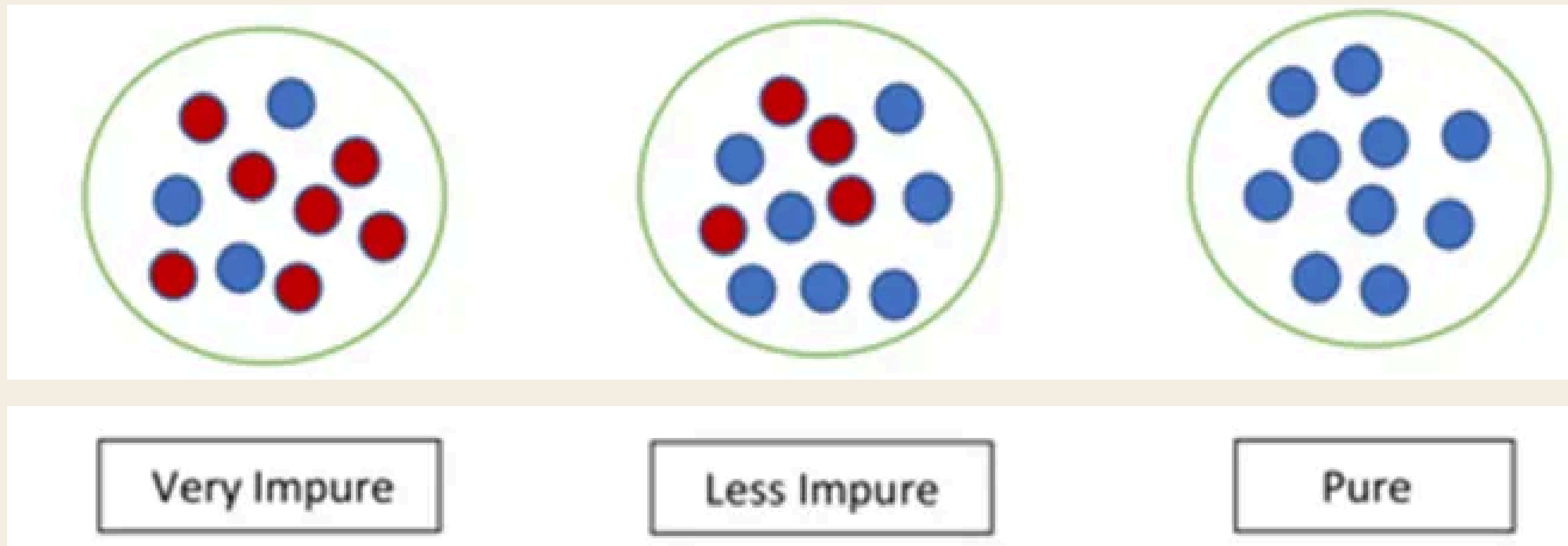
But how does we know / measure “improves purity” 🧐🧐

But first, what is purity?

The simple idea is:

- A node is pure if all observations belong to the same class.
- A node is impure if classes are mixed.

Let's try!



Measuring Impurity: Entropy

- Actually there are two ways to measure impurity: Gini and Entropy
- In this course, we can just focus on entropy (default of the course)

Entropy

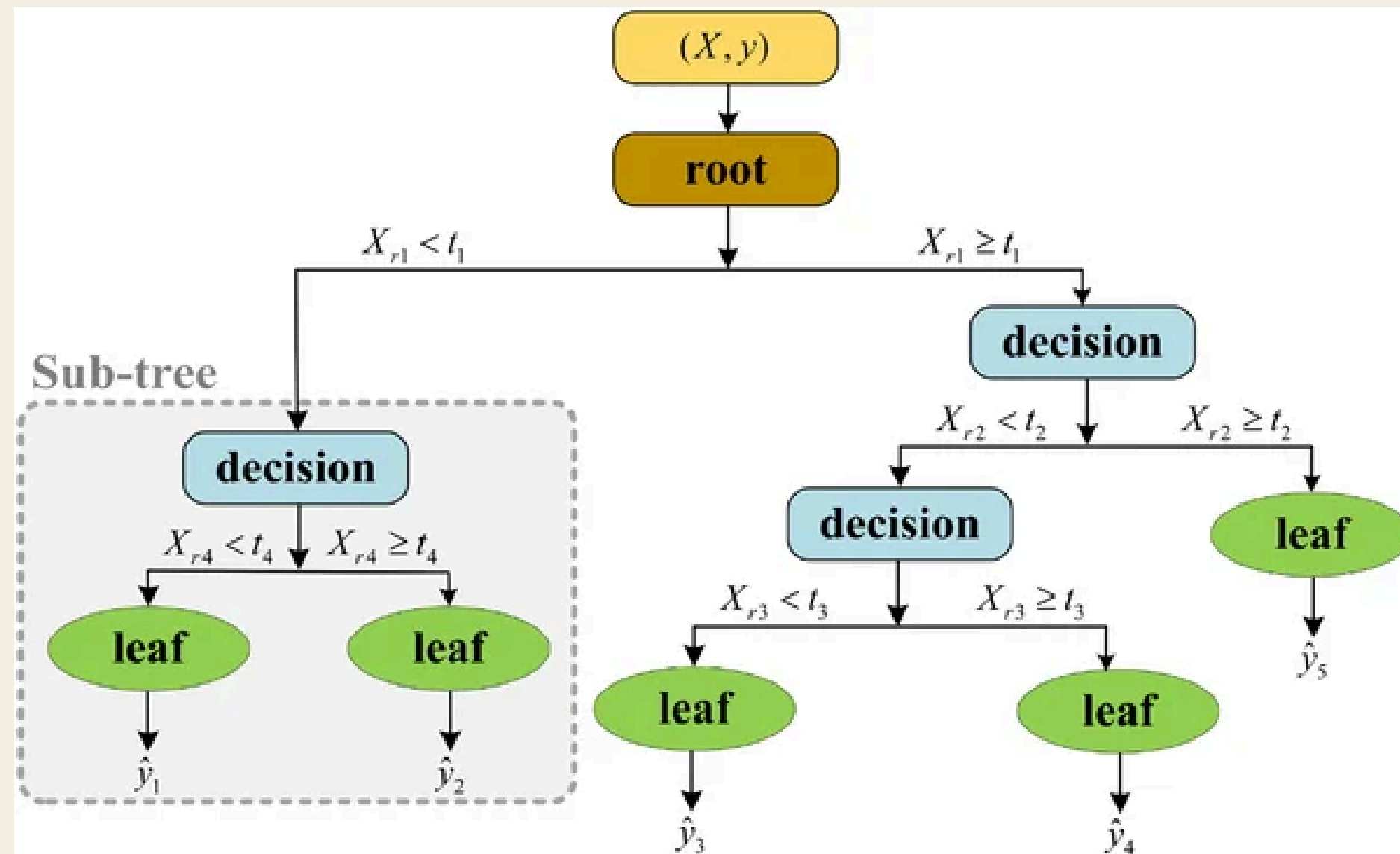
- Entropy measures how mixed a node is
- Pure node $\rightarrow$ Entropy = 0
- Evenly mixed (50/50) $\rightarrow$ Highest entropy
- The decision tree chooses the split that produces the **largest reduction in entropy** from parent to child nodes. *(So we want largest information gain)*

$$H = - \sum p \log_2 p$$

$$\text{Information Gain} = H(\text{parent}) - H(\text{children})$$

Quick Question: Is Decision Tree only for classification?

No! It can also be for regression. (Instead of class, just take mean of y)



But in this course,
we only focus on
classification!

(Extra) Read more: http://www.saedsayad.com/decision_tree_reg.htm

rpart() and rpart.plot() function in R

rpart(): to build the decision tree model

rpart.plot(): visualizing the decision tree created by rpart()

```
install.packages("rpart")  
install.packages("rpart.plot")  
  
# must load these two packages first!  
library(rpart)  
library(rpart.plot)
```

rmb to load package first before using rpart() and rpart.plot()!

rpart() and rpart.plot() function in R

so many parameters!

```
rpart(formula, data, weights, subset, na.action, method,  
      model, x, y, parms, control)
```

```
rpart.plot(x, type, extra, under, fallen.leaves,  
           digits, varlen, faclen, roundint,  
           cex, tweak, clip.facs, clip.right.labs,  
           snip, box.palette, shadow.col)
```

highlighted ones are the ones we mostly care about for this course

rpart() Parameters

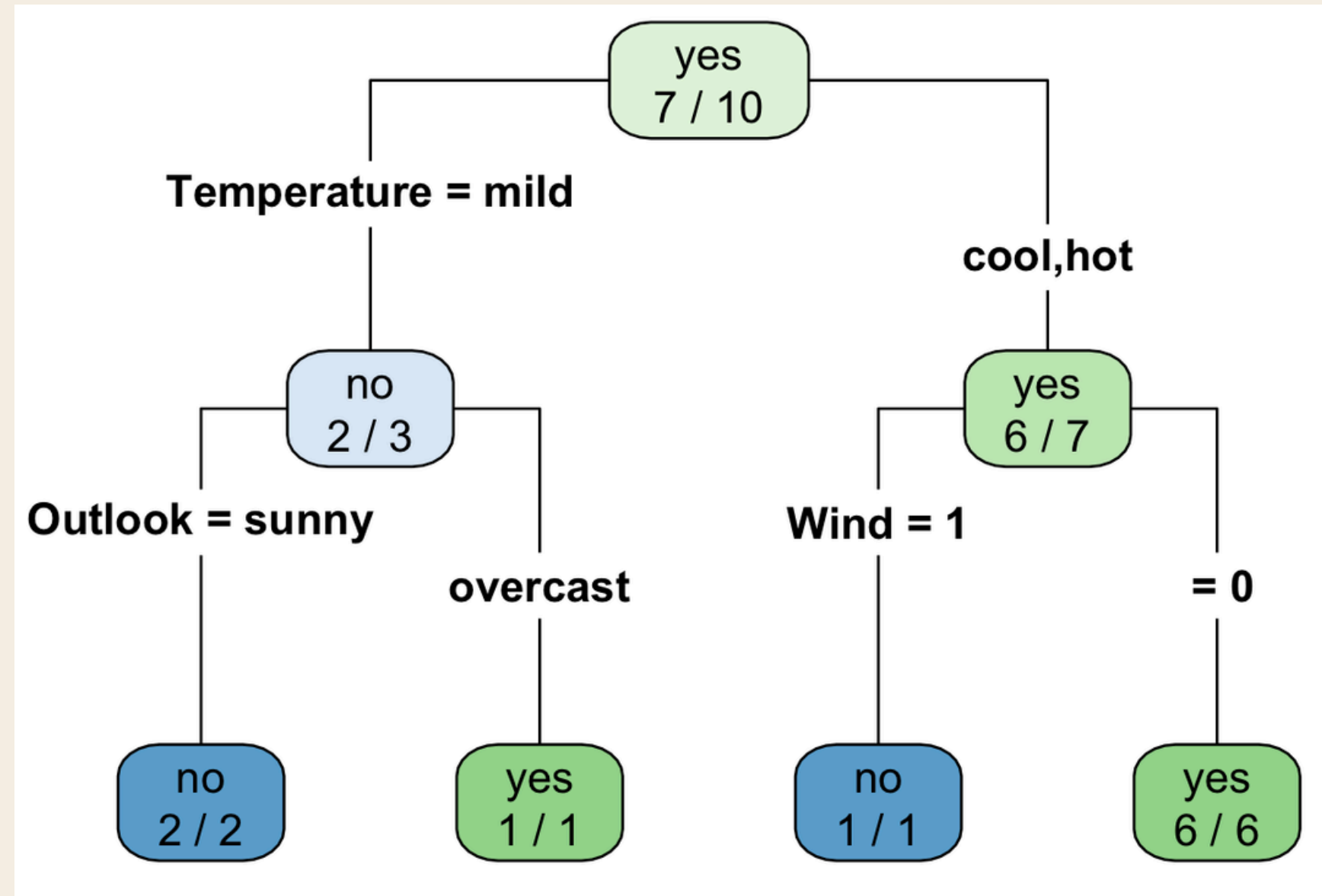
formula (required)	Specifies the response and predictor variables. Format: response ~ predictors Play ~ Outlook + Temperature + Humidity + Wind
data (required)	Data frame containing the variables in <i>formula</i> .
method (optional)	Specifies the type of tree. "class" → classification "anova" → regression "poisson" → count data "exp" → survival data
parms (optional)	Controls splitting behavior (mainly for classification). Format: parms = list(split = "information") Common: split = "gini" (default in R) split = "information" (entropy)
control (optional)	Controls tree growth. Format: control = rpart.control(...) Common: • cp → minimum improvement required to make a split • minsplit → minimum observations needed before splitting • minbucket → minimum observations allowed in a leaf • maxdepth → maximum depth of the tree

rpart.plot() Parameters

x (required)	The fitted rpart model to be plotted.
type	Controls how the tree layout and labels are displayed. 0 → splits shown above nodes 1 → splits inside nodes 2 → label all nodes 4 → clean classification-style layout (very common)
extra	Controls extra information displayed inside nodes. 0 → no extra info 1 → number of observations 2 → class probabilities 104 → predicted class + probability (very common)
varlen	Controls how many characters of variable names are shown in split labels. varlen = 0 → show full variable name
faclen	Controls how many characters of factor levels are shown in labels. faclen = 0 → show full level names
clip.right.labs	Whether long right-side labels are clipped (cut off) to avoid overlapping. TRUE → clip long labels FALSE → show full labels

Example of using `rpart()` and `rpart.plot()` for classification

```
fit <- rpart(Play ~ Outlook + Temperature + Humidity + Wind,  
  method = "class",  
  data = data,  
  control = rpart.control(minsplit=1),  
  parms = list(split='information'))  
  
rpart.plot(fit, type = 4, extra = 2)
```



Predicting using rpart() function - Classification

Returns the predicted class (majority class of the leaf node).

```
predict(fit, newdata, type = "class")
```

Returns the class probabilities for each observation

```
predict(fit, newdata, type = "prob")
```


How to evaluate classification decision tree?

our fav confusion matrix!

	Actually Positive (1)	Actually Negative (0)
Predicted Positive (1)	True Positives (TPs)	False Positives (FPs)
Predicted Negative (0)	False Negatives (FNs)	True Negatives (TNs)

Sometimes Negative and Positive will switch places. So please read carefully before answering!

Accuracy =

$$(TP + TN) / (TP + TN + FP + FN)$$

$$TPR (Recall) = TP / (TP + FN)$$

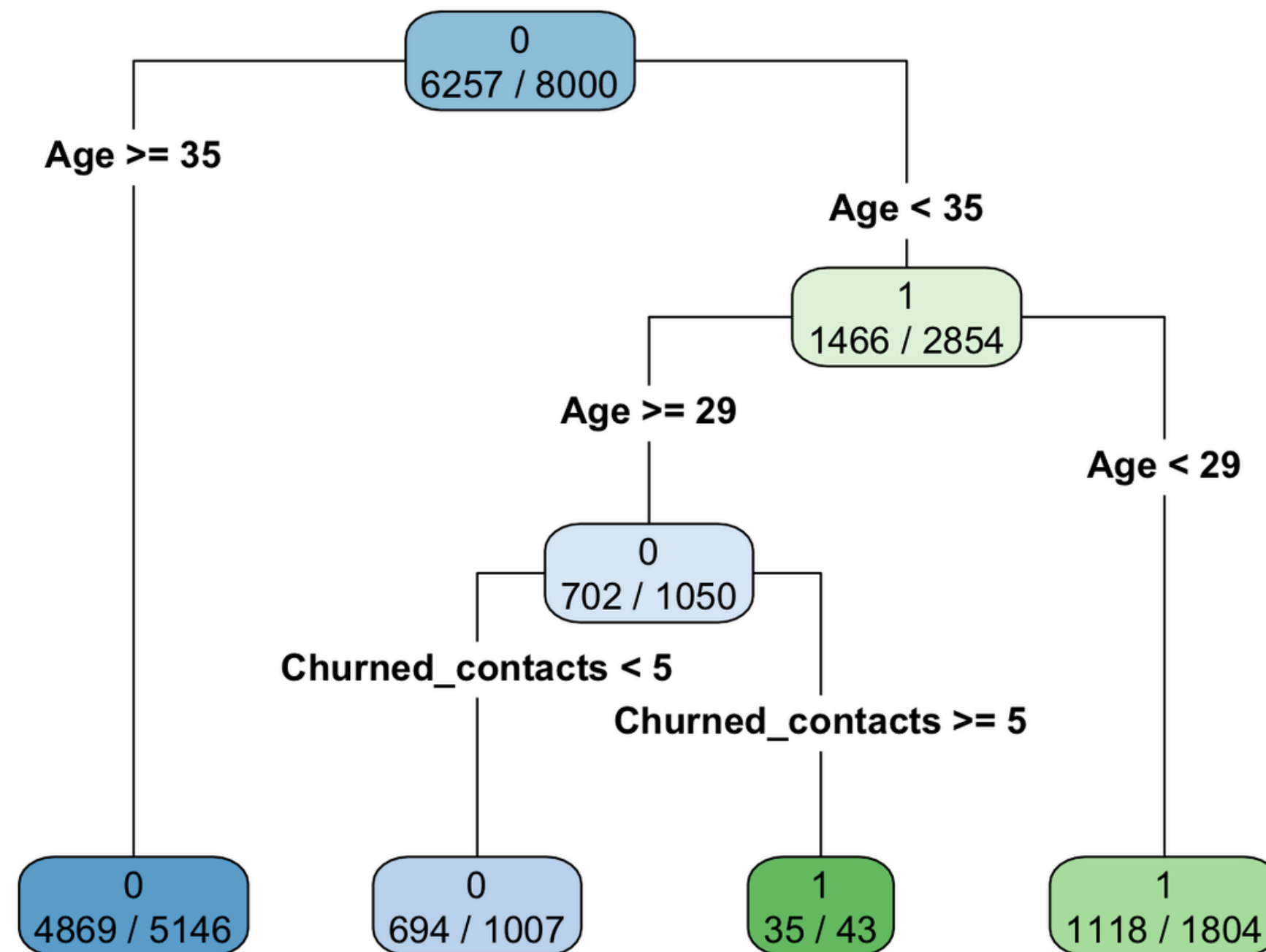
$$FPR = FP / (FP + TN)$$

$$FNR = FN / (TP + FN)$$

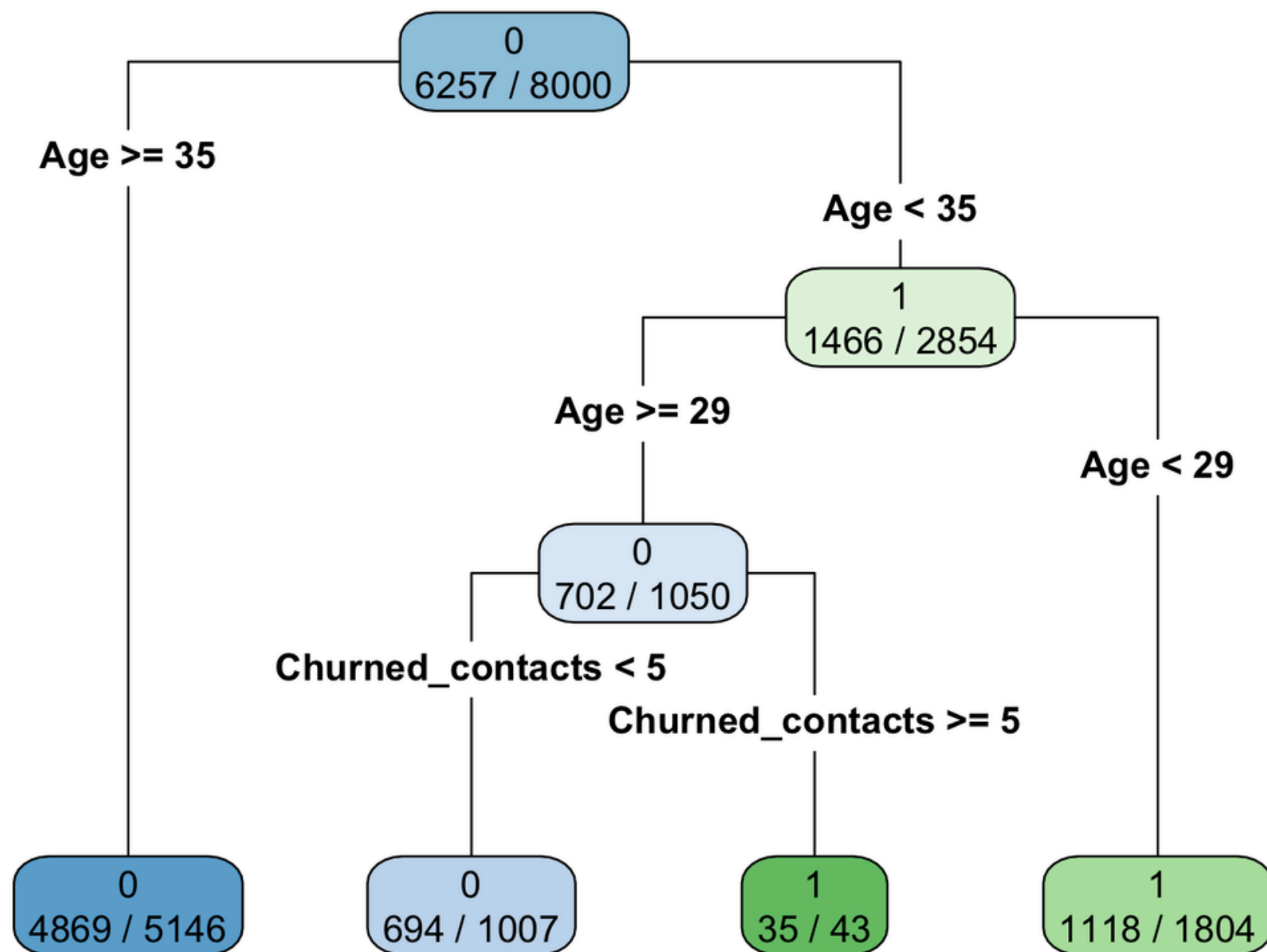
$$Precision = TP / (TP + FP)$$

Interpreting Decision Tree

How would you interpret this decision tree?



Interpreting Decision Tree



If Age $\geq 35 \rightarrow$ predict class 0

If Age < 35 AND Age $< 29 \rightarrow$ predict class 1

If Age < 35 AND Age ≥ 29 AND Churned_contacts $\geq 5 \rightarrow$ predict class 1

Root Node: Predicted class = 0

- 6257 out of 8000 observations in this node belong to class 0
-

In this course: **Features used closer to the root are generally more important.**

- as root split is chosen to give the largest impurity reduction at that moment (greedy choice)
- In this example, Age is clearly v imp (splits first and second)

dt\$variable.importance reflects a feature's overall contribution to the tree (total impurity reduction), so it may differ from simply looking at which feature appears closest to the root. (our course just looks at closest to root!!)

About setting seed

```
> set.seed(123)
> runif(3)
[1] 0.2875775 0.7883051 0.4089769
>
> set.seed(123)
> runif(3)
[1] 0.2875775 0.7883051 0.4089769
```

```
> set.seed(123)
> runif(2)    # uses first two numbers
[1] 0.2875775 0.7883051
> runif(2)    # uses next two numbers
[1] 0.4089769 0.8830174
```

`set.seed(123)`: Start the random number generator from position 123

About setting seed

Prof:

“For the test/exam (graded) then the question will specify when student should run the `set.seed()` function.

While for tutorial, it's suggested that student could run `set.seed()` before any `sample()` function.”

OFF-SITE QUESTIONS

Let's go to code!

Extra Materials

Outside this course, is Decision Tree often a good enough model? **No.**

Prone to overfitting and have high variance

Extra readings:

- **Bagging:** <https://medium.com/@damaniayesh/the-art-of-bagging-enhancing-decision-trees-with-ensemble-techniques-2786e4bffc27>
- **Bagging vs Boosting:** <https://medium.com/data-science-collective/bagging-vs-boosting-a-deep-dive-into-ensemble-learning-for-tree-based-models-b3a061da3e62>

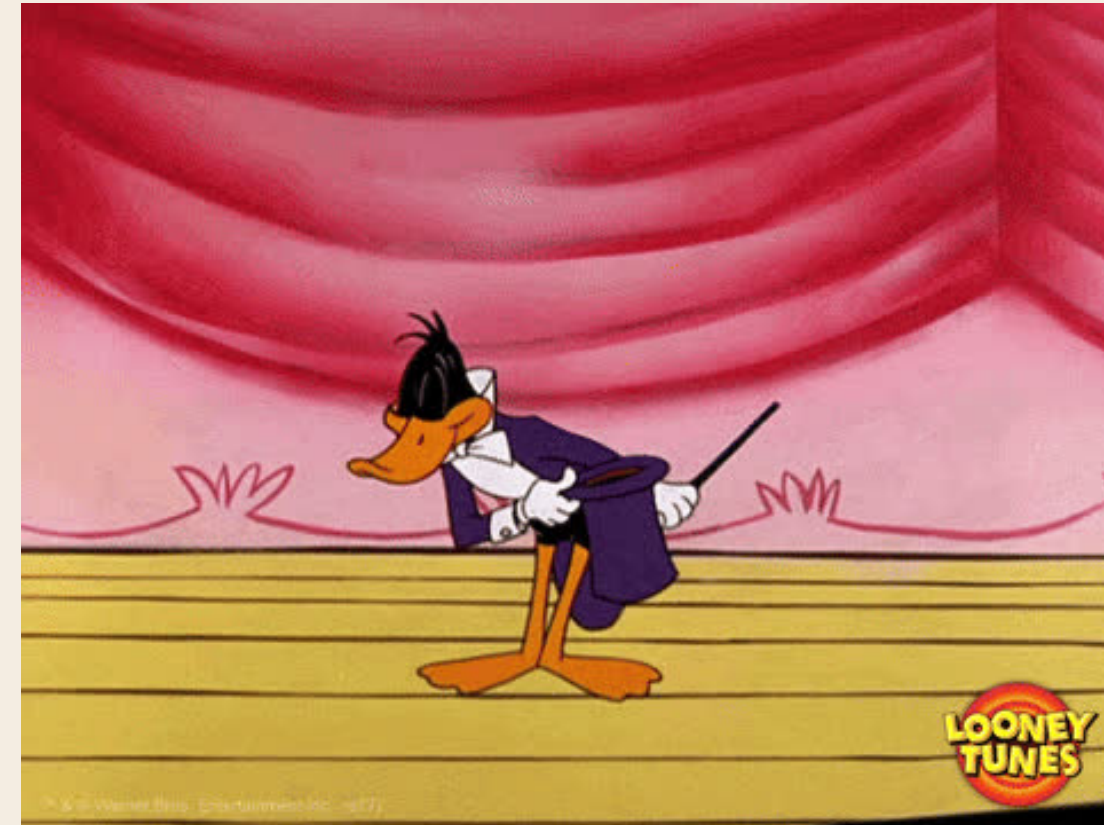
MIDTERMS PAST PAPERS



Thank you TA Jin
for compiling!

[https://drive.google.com/drive/folders/10MTJtBxbYA1
bRwHmmwGnICUSR5FsZ1ur?usp=sharing](https://drive.google.com/drive/folders/10MTJtBxbYA1bRwHmmwGnICUSR5FsZ1ur?usp=sharing)

**THANK YOU AND
GOOD LUCK FOR
EXAMS!**



Weekly Slides & Code will be shared on Github (★ star it for bookmark):

<https://github.com/kr7001/DSA1101-teaching-materials>